DMS613: Introduction to Mathematical Finance
Sub-hedging & Super-hedging

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1 An incomplete trinomial model

We begin with the example, because it shows immediately why incomplete markets are different from complete markets. Consider a one-period model with one stock and one bond. Let

$$S(0) = 100, \qquad 1 + r = 1.05,$$

and suppose the stock can take three possible values at time 1:

$$S(1) = \begin{cases} 130, & \omega_u, \\ 105, & \omega_m, \\ 80, & \omega_d. \end{cases}$$

The time-1 value of investing the current stock price in the bond is

$$(1 + r)S(0) = 105.$$

Since

$$80 < 105 < 130,$$

the stock return does not dominate the bond return and the bond return does not dominate the stock return. In this one-period one-stock model, this is the no-arbitrage condition.

A risk-neutral measure Q must satisfy

$$100 = \frac{1}{1.05} E^Q[S(1)].$$

Writing

$$Q(\omega_u) = q_u, \qquad Q(\omega_m) = q_m, \qquad Q(\omega_d) = q_d,$$

we need

$$q_u + q_m + q_d = 1, \qquad 130q_u + 105q_m + 80q_d = 105.$$

Substituting $q_m = 1 - q_u - q_d$ into the second equation gives

$$130q_u + 105(1 - q_u - q_d) + 80q_d = 105,$$

so

$$25q_u - 25q_d = 0, \qquad \text{and hence} \qquad q_u = q_d.$$

Thus the family of risk-neutral measures is

$$q_u = q_d = a, \qquad q_m = 1 - 2a, \qquad 0 \le a \le \frac{1}{2}.$$

Now consider a European call option with strike $K = 100$. Its payoff is

$$X = (S(1) - 100)^+ = \begin{cases} 30, & \omega_u, \\ 5, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

For the risk-neutral measure corresponding to a,

$$\frac{1}{1.05} E^Q[X] = \frac{1}{1.05}\big(30a + 5(1-2a) + 0 \cdot a\big) = \frac{1}{1.05}(5 + 20a).$$

As a ranges over $[0, 1/2]$, the discounted risk-neutral expectation ranges over

$$\left[\frac{5}{1.05}, \frac{15}{1.05}\right].$$

So the model is arbitrage-free, but the option does not have a single price determined by no-arbitrage alone. What is the no-arbitrage interval for the claim X?

The lower endpoint will be the subhedging price, and the upper endpoint will be the superhedging price. We first define these endpoints directly in terms of portfolios. Later, we state the theorem which says that these same endpoints are also obtained by minimizing and maximizing discounted risk-neutral expectations.

2 One-period market

We work in the one-period model developed earlier in the course. The state space is finite,

$$\Omega = \{\omega_1, \dots, \omega_n\},$$

and there are $d + 1$ traded assets. Their time-0 prices are collected in the vector

$$\bar{\pi} = (\pi_0, \pi_1, \dots, \pi_d) \in \mathbb{R}_+^{d+1},$$

and their time-1 payoffs are collected in the random vector

$$\bar{S} = (S^0, S^1, \dots, S^d).$$

As before, asset 0 is the risk-free bond, so that

$$\pi_0 = 1, \qquad S^0(\omega) = 1 + r \quad \text{for all } \omega \in \Omega,$$

where $r > -1$.

A portfolio is a vector

$$\bar{\xi} = (\xi_0, \xi_1, \dots, \xi_d) \in \mathbb{R}^{d+1}.$$

Its time-0 cost is

$$\bar{\pi} \cdot \bar{\xi},$$

and its time-1 payoff in state ω is

$$\bar{\xi} \cdot \bar{S}(\omega) = \sum_{i=0}^{d} \xi_i S^i(\omega).$$

We assume throughout this note that the original market is arbitrage-free. By the finite-state Fundamental Theorem of Asset Pricing proved earlier in the course, there exists at least one risk-neutral measure P^* such that

$$\pi_i = \frac{1}{1+r} E^*[S^i], \qquad i = 0, 1, \ldots, d.$$

If a contingent claim $X : \Omega \to \mathbb{R}$ is replicable, then there exists a portfolio $\bar{\xi}$ such that

$$X(\omega) = \bar{\xi} \cdot \bar{S}(\omega) \qquad \text{for every } \omega \in \Omega.$$

In that case, the law of one price forces the time-0 price of the claim to be $\bar{\pi} \cdot \bar{\xi}$.

In a complete market every claim is replicable, so every claim has a unique no-arbitrage price. In an incomplete market this need not happen. Instead of a single price, no-arbitrage usually gives a price interval.

3 Superhedging and subhedging prices

Suppose a new contingent claim with payoff X is introduced into the market. Think of X as the option payoff in the trinomial example above.

A seller of the claim is worried about the liability X at time 1. A conservative seller wants to construct a traded portfolio whose payoff is always at least X. This leads to superhedging.

Definition 3.1 (Superhedge). *A portfolio $\bar{\xi} \in \mathbb{R}^{d+1}$ is said to superhedge the contingent claim X if*

$$\bar{\xi} \cdot \bar{S}(\omega) \geq X(\omega) \qquad \textit{for every } \omega \in \Omega.$$

The superhedging price of X is

$$\pi_{\sup}(X) := \inf\big\{\bar{\pi} \cdot \bar{\xi} : \bar{\xi} \cdot \bar{S}(\omega) \geq X(\omega) \ \forall \omega \in \Omega\big\}.$$

So $\pi_{\sup}(X)$ is the cheapest cost at which the seller can fully cover the liability X in every state.

Similarly, a buyer of the claim can compare X with portfolios that are always below it. This leads to subhedging.

Definition 3.2 (Subhedge). *A portfolio $\bar{\eta} \in \mathbb{R}^{d+1}$ is said to subhedge the contingent claim X if*

$$\bar{\eta} \cdot \bar{S}(\omega) \leq X(\omega) \qquad \textit{for every } \omega \in \Omega.$$

The subhedging price of X is

$$\pi_{\text{sub}}(X) := \sup\big\{\bar{\pi} \cdot \bar{\eta} : \bar{\eta} \cdot \bar{S}(\omega) \leq X(\omega) \ \forall \omega \in \Omega\big\}.$$

So $\pi_{\text{sub}}(X)$ is the most expensive portfolio whose payoff is still dominated by X in every state.

Proposition 3.3. *For every contingent claim X,*

$$\pi_{\text{sub}}(X) = -\pi_{\sup}(-X).$$

Proof. A portfolio $\bar{\eta}$ satisfies

$$\bar{\eta} \cdot \bar{S} \leq X$$

if and only if $-\bar{\eta}$ satisfies

$$(-\bar{\eta}) \cdot \bar{S} \geq -X.$$

The cost of $-\bar{\eta}$ is $-\bar{\pi} \cdot \bar{\eta}$. Hence taking the supremum over subhedges of X is the same as taking minus the infimum over superhedges of $-X$. □

4 Why these prices give the endpoints of the no-arbitrage interval

The superhedging and subhedging prices are not arbitrary definitions. They give the two obvious arbitrage barriers.

Proposition 4.1. *Assume the original market is arbitrage-free. Then*

$$\pi_{\text{sub}}(X) \le \pi_{\text{sup}}(X).$$

Proof. Let $\bar{\eta}$ be a subhedge of X and let $\bar{\xi}$ be a superhedge of X. Then

$$\bar{\eta} \cdot \bar{S}(\omega) \le X(\omega) \le \bar{\xi} \cdot \bar{S}(\omega) \qquad \forall \omega \in \Omega.$$

Therefore

$$(\bar{\xi} - \bar{\eta}) \cdot \bar{S}(\omega) \ge 0 \qquad \forall \omega \in \Omega.$$

If

$$\bar{\pi} \cdot (\bar{\xi} - \bar{\eta}) < 0,$$

then the portfolio $\bar{\xi} - \bar{\eta}$ would have negative initial cost and nonnegative terminal payoff in every state. This would be an arbitrage opportunity. Hence

$$\bar{\pi} \cdot \bar{\eta} \le \bar{\pi} \cdot \bar{\xi}.$$

Taking the supremum over all subhedges and the infimum over all superhedges gives

$$\pi_{\text{sub}}(X) \le \pi_{\text{sup}}(X).$$

□

Proposition 4.2 (Replicable claims)**.** *If X is replicable by a portfolio $\bar{\xi}$, then*

$$\pi_{\text{sub}}(X) = \pi_{\text{sup}}(X) = \bar{\pi} \cdot \bar{\xi}.$$

Proof. If X is replicated by $\bar{\xi}$, then

$$\bar{\xi} \cdot \bar{S}(\omega) = X(\omega) \qquad \forall \omega \in \Omega.$$

So $\bar{\xi}$ is both a superhedge and a subhedge. Therefore

$$\pi_{\text{sub}}(X) \le \bar{\pi} \cdot \bar{\xi} \le \pi_{\text{sup}}(X).$$

The previous proposition gives the reverse ordering between $\pi_{\text{sub}}(X)$ and $\pi_{\text{sup}}(X)$, so equality holds throughout. □

Now suppose the new claim is quoted at time-0 price p.

Proposition 4.3. *Assume the original market is arbitrage-free. If p is an arbitrage-free price for the claim X, then*

$$\pi_{\text{sub}}(X) \le p \le \pi_{\text{sup}}(X).$$

Proof. Suppose first that $p > \pi_{\text{sup}}(X)$. Then, by definition of the infimum, there exists a superhedging portfolio $\bar{\xi}$ such that

$$\bar{\pi} \cdot \bar{\xi} < p, \qquad \bar{\xi} \cdot \bar{S}(\omega) \ge X(\omega) \quad \forall \omega \in \Omega.$$

Sell one unit of the claim for p, and use part of the proceeds to buy the superhedging portfolio $\bar{\xi}$. The initial net cash inflow is

$$p - \bar{\pi} \cdot \bar{\xi} > 0.$$

At time 1, the payoff of the hedge minus the liability on the short claim is

$$\bar{\xi} \cdot \bar{S}(\omega) - X(\omega) \geq 0 \qquad \forall \omega \in \Omega.$$

Thus we have an arbitrage opportunity. Therefore an arbitrage-free price cannot satisfy $p > \pi_{\mathrm{sup}}(X)$.

Similarly, suppose $p < \pi_{\mathrm{sub}}(X)$. Then there exists a subhedging portfolio $\bar{\eta}$ such that

$$\bar{\pi} \cdot \bar{\eta} > p, \qquad \bar{\eta} \cdot \bar{S}(\omega) \leq X(\omega) \quad \forall \omega \in \Omega.$$

Buy one unit of the claim for p and short the portfolio $\bar{\eta}$. The initial net cash inflow is

$$\bar{\pi} \cdot \bar{\eta} - p > 0.$$

At time 1, the payoff of the long claim minus the payoff owed on the short portfolio is

$$X(\omega) - \bar{\eta} \cdot \bar{S}(\omega) \geq 0 \qquad \forall \omega \in \Omega.$$

Again we obtain an arbitrage opportunity. Hence an arbitrage-free price cannot satisfy $p < \pi_{\mathrm{sub}}(X)$.

□

5 Computing superhedges and subhedges by linear programming

The definitions above are statewise inequalities. In a finite-state model, statewise inequalities are just finitely many linear inequalities. Therefore superhedging and subhedging are linear programming problems.

Let

$$x_j := X(\omega_j), \qquad x = (x_1, \dots, x_n)^\top \in \mathbb{R}^n.$$

Define the $n \times (d+1)$ payoff matrix A by

$$A_{ji} = S^i(\omega_j), \qquad j = 1, \dots, n, \quad i = 0, \dots, d.$$

Then the vector of portfolio payoffs across states is

$$A\bar{\xi} = \begin{pmatrix} \bar{\xi} \cdot \bar{S}(\omega_1) \\ \vdots \\ \bar{\xi} \cdot \bar{S}(\omega_n) \end{pmatrix}.$$

5.1 The superhedging linear program

The condition that $\bar{\xi}$ superhedges X is

$$A\bar{\xi} \geq x,$$

where the inequality is componentwise. Hence

$$\pi_{\mathrm{sup}}(X) = \min_{\bar{\xi} \in \mathbb{R}^{d+1}} \bar{\pi}^\top \bar{\xi} \qquad \text{subject to} \qquad A\bar{\xi} \geq x.$$

This is the primal superhedging linear program.

The variables are the portfolio holdings. The objective is the initial cost. The constraints require the portfolio payoff to dominate the claim in every state.

5.2 The subhedging linear program

Similarly, the condition that $\bar{\eta}$ subhedges X is

$$A\bar{\eta} \le x.$$

Hence

$$\pi_{\text{sub}}(X) = \max_{\bar{\eta}\in\mathbb{R}^{d+1}} \bar{\pi}^\top \bar{\eta} \qquad \text{subject to} \qquad A\bar{\eta} \le x.$$

This is the subhedging linear program.

Equivalently, one can compute it from the superhedging problem using

$$\pi_{\text{sub}}(X) = -\pi_{\text{sup}}(-X).$$

5.3 LP computation in the trinomial example

Return to the trinomial model of Section 1. A portfolio consists of β units of the bond and Δ units of the stock. Its time-0 cost is

$$\beta + 100\Delta,$$

and its time-1 payoff is

$$1.05\beta + \Delta S(1).$$

The call payoff is

$$X(\omega_u) = 30, \qquad X(\omega_m) = 5, \qquad X(\omega_d) = 0.$$

The superhedging linear program is

$$\begin{aligned} \text{minimize} \quad & \beta + 100\Delta \\ \text{subject to} \quad & 1.05\beta + 130\Delta \ge 30, \\ & 1.05\beta + 105\Delta \ge 5, \\ & 1.05\beta + 80\Delta \ge 0. \end{aligned}$$

The optimal superhedge is obtained by taking the line through the down and up payoffs:

$$1.05\beta + 80\Delta = 0, \qquad 1.05\beta + 130\Delta = 30.$$

Solving gives

$$\Delta = \frac{30}{50} = 0.6, \qquad 1.05\beta = -48, \qquad \beta = -\frac{48}{1.05}.$$

The payoff of this portfolio is

state	ω_d	ω_m	ω_u
$S(1)$	80	105	130
portfolio payoff	0	15	30
X	0	5	30

Thus it superhedges X. Its cost is

$$\beta + 100\Delta = -\frac{48}{1.05} + 60 = \frac{15}{1.05}.$$

Therefore

$$\pi_{\text{sup}}(X) = \frac{15}{1.05}.$$

The subhedging linear program is

$$\begin{aligned} \text{maximize} \quad & \beta + 100\Delta \\ \text{subject to} \quad & 1.05\beta + 130\Delta \le 30, \\ & 1.05\beta + 105\Delta \le 5, \\ & 1.05\beta + 80\Delta \le 0. \end{aligned}$$

Since the middle-state payoff constraint gives

$$1.05\beta + 105\Delta \le 5,$$

and

$$\beta + 100\Delta = \frac{1}{1.05}\big(1.05\beta + 105\Delta\big),$$

we immediately get

$$\beta + 100\Delta \le \frac{5}{1.05}.$$

This upper bound is achieved, for example, by taking the line through the down and middle payoffs:

$$1.05\beta + 80\Delta = 0, \qquad 1.05\beta + 105\Delta = 5.$$

Solving gives

$$\Delta = \frac{5}{25} = 0.2, \qquad 1.05\beta = -16, \qquad \beta = -\frac{16}{1.05}.$$

The payoff of this portfolio is

state	ω_d	ω_m	ω_u
$S(1)$	80	105	130
portfolio payoff	0	5	10
X	0	5	30

Thus it subhedges X, and its cost is

$$\beta + 100\Delta = -\frac{16}{1.05} + 20 = \frac{5}{1.05}.$$

Therefore

$$\pi_{\text{sub}}(X) = \frac{5}{1.05}.$$

So the hedging computation gives

$$\pi_{\text{sub}}(X) = \frac{5}{1.05}, \qquad \pi_{\text{sup}}(X) = \frac{15}{1.05}.$$

This matches the interval obtained earlier from the family of risk-neutral measures.

6 Superhedging-subhedging theorem

We now record the general result that connects the hedging prices computed above with risk-neutral expectations. At this stage, we only state the theorem and do not prove it.

Definition 6.1 (Risk-neutral measures). *Let $\overline{\mathcal{M}}$ denote the set of all probability measures Q on Ω such that*

$$Q(\omega_j) \geq 0, \qquad \sum_{j=1}^{n} Q(\omega_j) = 1,$$

and

$$\pi_i = \frac{1}{1+r} E^Q[S^i], \qquad i = 0, 1, \ldots, d.$$

Thus $\overline{\mathcal{M}}$ is the set of all risk-neutral measures for the one-period market.

Theorem 6.1 (Superhedging and subhedging prices). *Assume the original market is arbitrage-free. Then, for every contingent claim $X : \Omega \to \mathbb{R}$,*

$$\pi_{\text{sup}}(X) = \frac{1}{1+r} \sup_{Q \in \overline{\mathcal{M}}} E^Q[X],$$

and

$$\pi_{\text{sub}}(X) = \frac{1}{1+r} \inf_{Q \in \overline{\mathcal{M}}} E^Q[X].$$

Equivalently, the no-arbitrage price interval for X is

$$\left[\frac{1}{1+r} \inf_{Q \in \overline{\mathcal{M}}} E^Q[X], \frac{1}{1+r} \sup_{Q \in \overline{\mathcal{M}}} E^Q[X] \right].$$

Remark 1. *The theorem says that the lowest and highest arbitrage-free prices are obtained by taking the smallest and largest discounted risk-neutral expectations. In a complete market this interval collapses to a single point. In an incomplete market it may be a genuine interval.*

Exercises

Exercises

1. Consider the one-period trinomial model from the notes:

$$S(0) = 100, \qquad 1 + r = 1.05,$$

and

$$S(1) = \begin{cases} 130, & \omega_u, \\ 105, & \omega_m, \\ 80, & \omega_d. \end{cases}$$

Let $Q(\omega_u) = q_u$, $Q(\omega_m) = q_m$, and $Q(\omega_d) = q_d$.

1. Write down the equations that q_u, q_m, q_d must satisfy for Q to be risk-neutral.
2. Show that every risk-neutral measure is of the form

$$q_u = q_d = a, \qquad q_m = 1 - 2a, \qquad 0 \leq a \leq \frac{1}{2}.$$

3. Explain why the market is incomplete.

2. In the same trinomial model, consider the payoff

$$Y = \begin{cases} 20, & \omega_u, \\ 10, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

   1. Compute
   $$\frac{1}{1.05} E^Q[Y]$$
   for the risk-neutral measure indexed by a.
   2. Find the smallest and largest possible discounted risk-neutral expectations.
   3. Hence identify the subhedging and superhedging prices of Y.
   4. Is Y replicable? Justify your answer.

3. In the trinomial model, consider the digital payoff

$$D = \begin{cases} 1, & \omega_u, \\ 0, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

   1. Compute the interval of discounted risk-neutral expectations of D.
   2. Find $\pi_{\text{sub}}(D)$ and $\pi_{\text{sup}}(D)$.
   3. Set up the superhedging linear program for D.
   4. Set up the subhedging linear program for D.

4. In the trinomial call option example from the notes, the payoff is

$$X = (S(1) - 100)^+ = \begin{cases} 30, & \omega_u, \\ 5, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

   The notes show that
   $$\pi_{\text{sub}}(X) = \frac{5}{1.05}, \qquad \pi_{\text{sup}}(X) = \frac{15}{1.05}.$$

   For each of the following quoted prices, decide whether the price is arbitrage-free. If it is not arbitrage-free, describe the arbitrage strategy.

   1. $p = 4$.
   2. $p = 10$.
   3. $p = 16$.

5. Again consider the call payoff

$$X = \begin{cases} 30, & \omega_u, \\ 5, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

   A portfolio consists of β units of the bond and Δ units of the stock.

   1. Write the three inequalities that say the portfolio superhedges X.
   2. Solve the two equations

$$1.05\beta + 80\Delta = 0, \qquad 1.05\beta + 130\Delta = 30.$$

   3. Check that the resulting portfolio payoff dominates X in all three states.
   4. Compute its initial cost.
   5. Explain why this cost is the superhedging price.

6. For the same call payoff X, consider the portfolio obtained by solving

$$1.05\beta + 80\Delta = 0, \qquad 1.05\beta + 105\Delta = 5.$$

   1. Solve for β and Δ.
   2. Compute the portfolio payoff in all three states.
   3. Check that this payoff is below X in every state.
   4. Compute the initial cost.
   5. Explain why this cost is the subhedging price.

7. In the trinomial model, a claim Z is replicable if there exist $\beta, \Delta \in \mathbb{R}$ such that

$$1.05\beta + \Delta S(1) = Z$$

   in all three states.

   For each payoff below, determine whether it is replicable. If it is replicable, find the replicating portfolio and its price. If it is not replicable, explain why.

$$Z_1 = \begin{cases} 30, & \omega_u, \\ 15, & \omega_m, \\ 0, & \omega_d, \end{cases} \qquad Z_2 = \begin{cases} 30, & \omega_u, \\ 5, & \omega_m, \\ 0, & \omega_d, \end{cases} \qquad Z_3 = \begin{cases} 10, & \omega_u, \\ 5, & \omega_m, \\ 0, & \omega_d. \end{cases}$$

8. Consider a one-period finite-state market with states

$$\Omega = \{\omega_1, \dots, \omega_n\}.$$

   Let A be the payoff matrix whose j-th row gives the payoffs of the traded assets in state ω_j, and let $x_j = X(\omega_j)$.

   1. Write the superhedging linear program for X.
   2. Write the subhedging linear program for X.
   3. Explain the financial meaning of the variables in each program.
   4. Explain why there is one constraint for each state of the world.